

National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

FACULTY OF COMPUTING

CS-415: Mobile Application Development

Class: BESE-13AB

Lab 06: Navigation, Passing information and Alert Dialog, State Management

CLO-2: Design data-driven Mobile Interfaces using advanced design and development techniques

CLO-4: Display proficiency using modern tools and framework usage by developing interactive mobile applications

Date: 4th Mar 2026

Time: 12:00 - 14:25

Instructor: Ms. Naema Asif

Lab 06: Navigation, Passing information and Alert Dialog, State Management

Introduction:

This lab will focus on getting familiar with advanced concepts in material design, navigation and adding interactivity to the app and state management. Students will learn how to move between different screens using Flutter navigation techniques and how to pass data efficiently between them. The lab also introduces Alert Dialog and Snack Bar components to improve user interaction and feedback within the application. In addition, learners will understand how proper state management helps maintain consistency and improves the overall user experience in mobile applications.

Objectives:

In this lab, you will practice and develop understanding of the following concepts:

- Adding interactivity in Flutter applications using AlertDialog for confirmation and user response handling
- Implementing navigation between multiple screens using Flutter Navigator
- Passing data from one screen to another and returning data back to the previous screen
- Displaying SnackBar messages for instant feedback and better user experience
- Managing the state of selected and unselected items dynamically within the application
- Understanding how state changes affect the user interface and application behavior
- Improving user experience by handling delete actions with confirmation dialogs
- Building responsive and user-friendly mobile applications using modern Flutter development practices
- Applying Material Design principles to create clean and interactive app interfaces
- Strengthening problem-solving skills through hands-on implementation of real-world tasks.

Description :

This lab is designed to strengthen the understanding of interactive mobile application development using Flutter. Students will work with an existing project and enhance it by implementing navigation between screens and passing information correctly. They will also learn how to use Alert Dialog for confirmation actions and Snack Bar for instant user feedback. Special focus is given to state management so that UI updates correctly when items are selected, deleted, or modified. The tasks encourage practical implementation of user-centered design principles and improve problem-solving skills. By completing this lab, students will gain hands-on experience in building responsive and efficient Flutter applications. These concepts are essential for developing real-world mobile apps with better usability and maintainability.

Tools/Software Requirements

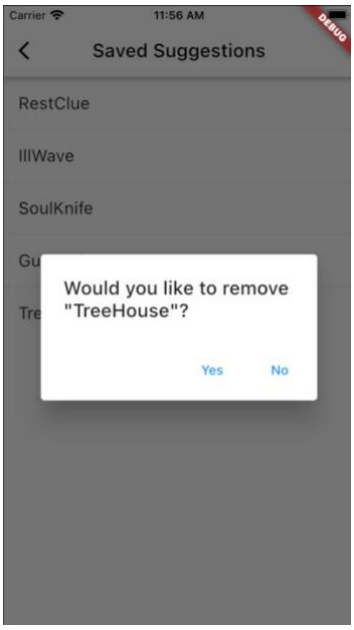
- Android Studio Ladybug Feature Drop | 2024.2.2
- Android Emulator (35.3.11) Stable *(Optional, depends on system's requirement)*
- Visual Studio Code (1.97.2 - Latest)
- Flutter Windows Android SDK 3.29.0 (stable)
- Flutter Lab Online

Lab Task-1:

This task requires you to use the previous codebase and modify it to improve user interaction and state handling in the **Saved Suggestions** screen. The main purpose of this task is to implement a confirmation process before deleting any saved item so that accidental deletion can be avoided. Students will learn how to use **AlertDialog**, manage navigation between screens, and update the application state correctly.

Additional Requirements and Explanation:

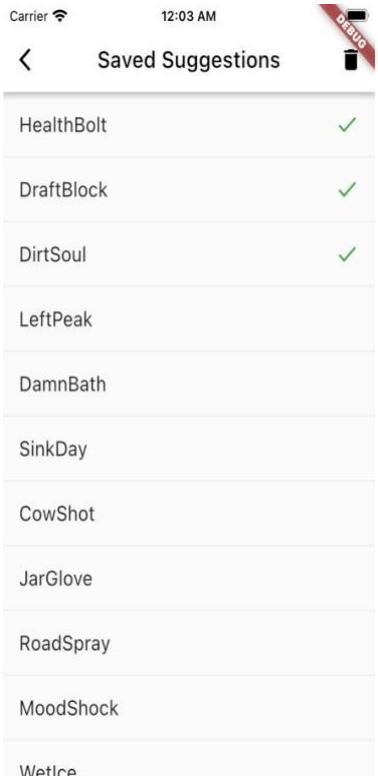
- Open the existing Flutter project created in the previous lab and locate the **Saved Suggestions** screen
- Identify the list of saved suggestions where the user can tap on any saved item
- Modify the tap functionality so that tapping an item does not delete it immediately
- Instead, display a confirmation **Alert Dialog** asking the user whether they want to remove the selected item
- The dialog should contain two buttons: **Yes** and **No**
- The Alert Dialog should be modal, meaning tapping outside the dialog should not close it
- If the user taps **Yes**, the selected item should be removed from the Saved Suggestions list
- After removing the item, the dialog should close automatically
- The app should then navigate the user back to the previous screen
- Also ensure that the same item is marked as **unselected** in the **Startup Name Generator** screen
- This ensures proper state synchronization between both screens
- If the user taps **No**, the dialog should simply close without making any changes
- No item should be removed when the user selects **No**
- Test the functionality multiple times to ensure the list updates correctly
- Verify that deleted items do not remain selected when returning to the previous screen
- Make sure the user interface remains smooth and responsive during these interactions



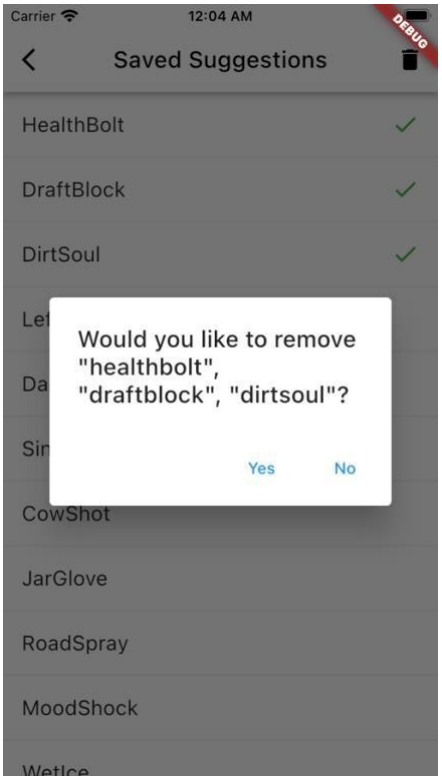
Lab Task-2:

Using the same above code base, make below changes:

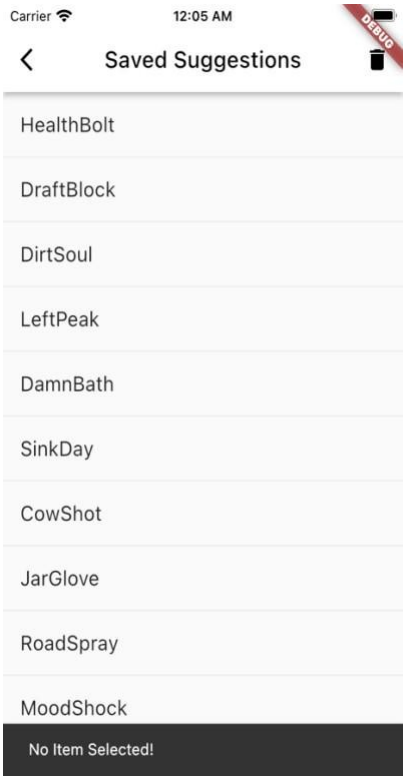
- Modify the code in "Saved Suggestions" screen to meet following requirements
- User should be able to select / deselect an item from the list
- "Selected" item should have a checkmark icon on the right side (screenshot 1)
- Tapping a "selected" item should deselect it and remove the checkmark (and vice versa)
- User should be able to tap the "Delete" button from AppBar to see a confirmation dialog to delete the "selected" items from the list (screenshot 2)
- Tapping the "Yes" button should:
- Remove the selected items from the "Saved Suggestions" list
    - Dismiss the dialog
    - Take user back to the previous screen
    - Also make sure that state of removed items is also set to "unselected" in the "Startup Name Generator" List
- Tapping the "No" button should just dismiss the dialog
- Tapping outside the AlertDialog should not dismiss the dialog
- The AlertDialog should only be visible when at least one item is selected from the list
- In case the User taps the "Delete' button without selecting an item from the list, display a SnackBar at the bottom of the screen with the message "No Item Selected" (screenshot 3)



Screenshot 1: Selected/unselected items



Screenshot 2: AlertDialog



Screenshot 3: Snackbar

Useful Resources:

Below is the list of useful resources to help you in solving above tasks:

- Navigate and passing data between screens
- Returning data from a screen
- Flutter AlertDialog class
- Flutter SnackBar
- State Management

Deliverable:

1. Compile a single word document by filling in the solution part and submit this Word file on LMS before deadline.
2. Submit all your code files on LMS.
3. Include screenshots of the program outputs.
4. Submit your Lab Word File and code files separately on submission link.

Lab Rubrics

	Lab 06 (Navigation, Passing information and Alert Dialog, State Management)				
Sr. No.	**Assessment**	**Unacceptable**	**Does Not Meet Expectations**	**Meets Expectations**	**Exceeds Expectations**
		(0 Marks)	**(1-2 Marks)**	**(3-4 Marks)**	**(5 Marks)**
1	**CLO-2:** Design data-driven Mobile Interfaces using advanced design and development techniques	The student did not submit any work. OR The student plagiarized the solution and/or used unfair means.	The student is unable to properly implement navigation, AlertDialog, SnackBar, and state management. The program has incorrect outputs, missing logic, or incomplete implementation. No proper screenshots are provided, or the submission	The student properly demonstrates navigation, data passing, Alert Dialog, and Snack Bar implementation. The program works correctly with proper state handling and outputs. At least one screenshot is provided with only minor file naming or late submission issues.	The student demonstrates a strong understanding of navigation flow, passing data between screens, Alert Dialog confirmation handling, Snack Bar usage, and state synchronization between screens. The program executes perfectly, handles all user interactions smoothly, and generates accurate outputs with proper UI behavior. Provided accurate screenshots

			is late/incorrectly named.		including all required outputs. Submitted with the correct file name and on time.
2	**CLO-4:** Display proficiency using modern tools and framework usage by developing interactive mobile applications	The student did not perform the lab activity during the lab session. OR The student used unfair means, copied the solution, or was unable to demonstrate the task using modern tools and frameworks during the lab assessment.	The student struggles with basic (Android Studio , Xcode & VS) controls, often requiring assistance to perform common tasks. They have difficulty navigating the IDE, using keyboard shortcuts, and managing the interface.	The student has a basic command of (Android Studio , Xcode & VS) controls, but may need guidance for more complex tasks. They can perform common actions but may do so slowly or with occasional errors.	The student demonstrates proficient use of (Android Studio , Xcode & VS) controls. They can efficiently navigate the IDE, utilize keyboard shortcuts effectively, and manage the interface without significant issues.
Obtained Marks:					